

VideoRoPE: What Makes for Good Video Rotary Position Embedding?

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Code

- <https://github.com/WiseInn570/VideoRoPE>

Title

VideoRoPE

Abstract & Introduction

- Extending 1D RoPE to video is challenging due to **spatio-temporal** structure.
- Identify four key traits: **3D structure**, **frequency allocation**, **spatial symmetry**, **temporal index scaling**.
- Introduce **V-NIAH-D** (distractors) showing prior variants misled by periodic oscillations.
- Propose VideoRoPE: **Low-frequency time**, **Diagonal layout**, Adjustable temporal spacing.

Overview

Why new RoPE for video?

What makes good Video RoPE?

- **2D/3D structure**: preserve t, x, y without flattening artifacts.
- **Frequency allocation**: assign low freq to time to avoid aliasing.
- **Spatial symmetry**: balanced text–video–text distances.
- **Temporal scaling**: decouple frame vs. text granularity.

Design goals

4 requirements

Experimental Setup

Data

- LLaVA-Video-178k subset (~**1.3M** QA pairs).
- Videos $\leq 2\text{--}3$ minutes; sampled at **2 fps**.

Models & Training

- Backbone: Qwen2-VL-7B vision; Qwen2-7B LLM.
- Window **8192** (train); vLLM for >32k inference.
- LR 1e-5, cosine, warmup 1e-2, batch 128; 704 A100 hours.

6
Benchmarks

2 fps
Sampling

128
Max frames

Setup

Data & training

Metrics

- **LongVideoBench / MLVU**: average accuracy (%) across tasks and context lengths (8k/16k/32k/64k).
- **Video-MME**: average accuracy (%) over evaluation categories and lengths.
- **V-NIAH / V-NIAH-D**: retrieval **accuracy** (%) averaged over haystack lengths and frame depths.
- **VideoHallucener**: category-wise scores (OR, T, SD, F, NF) reported as **accuracy** (%), averaged per category.

Analysis (Condensed)

- Lower dims (high freq) → **local**; higher dims (low freq) → **long-range**.
- M-RoPE gives high freq to t → temporal **aliasing**; distractor sensitivity in V-NIAH-D.
- **Spatial symmetry** reduces order bias; **temporal scaling** matches frame/text granularity.

VideoRoPE: Design

- **3D structure** over (t,x,y), preserving spatio-temporal relations.
- **Low-frequency Temporal Allocation** (t in higher dims; x/y interleaved in lower dims; x=48,y=48,t=32).
- **Diagonal layout** keeps symmetry and vanilla-like index growth.
- **Adjustable Temporal Spacing** with scaling factor δ for temporal indices.

Qualitative: Retrieval (V-NIAH / V-NIAH-D)

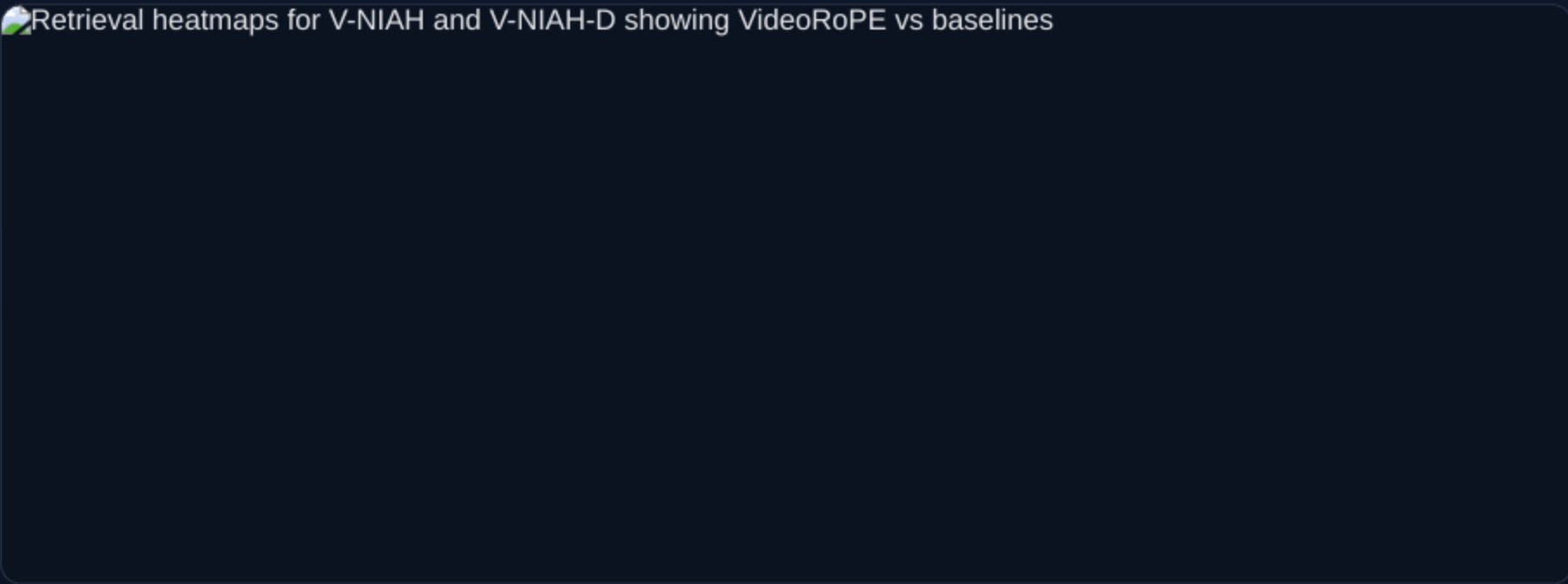


Figure 7. Color gradient from green to red shows needle retrieval performance (V-NIAH and V-NIAH-D).

Notice **VideoRoPE** remains **robust to distractors** while baselines degrade on V-NIAH-D.

Qualitative: Attention Decomposition

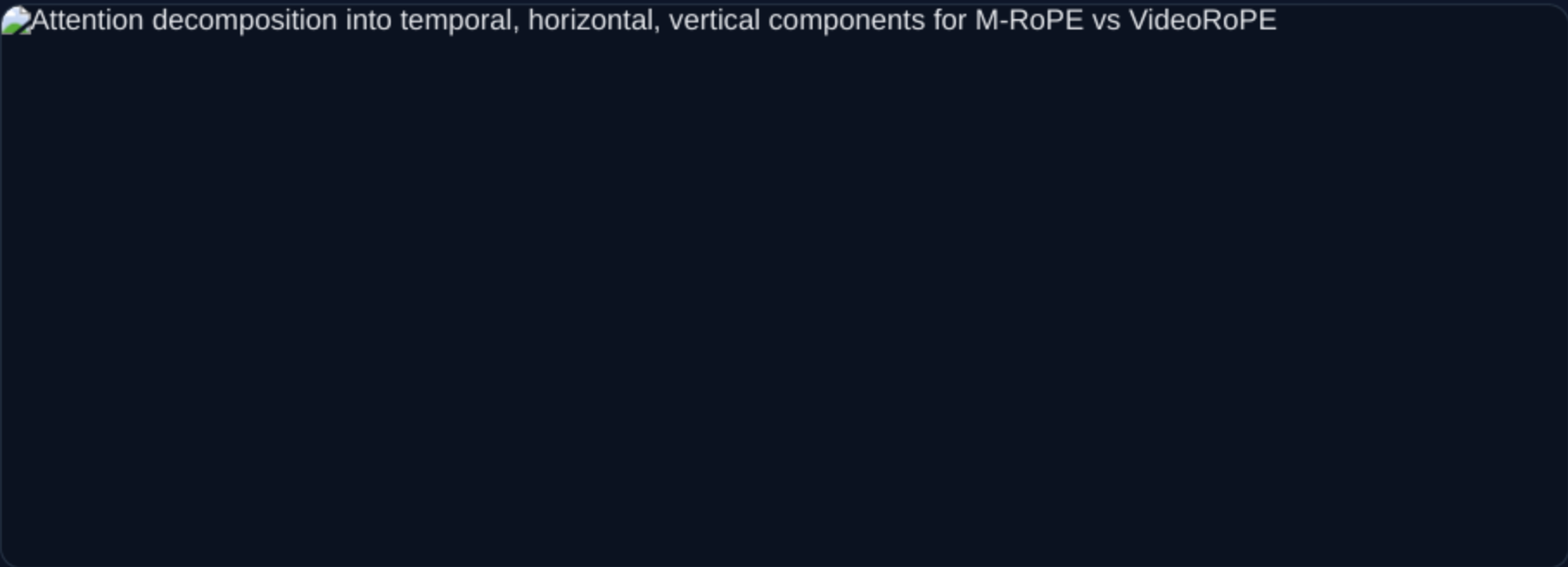


Figure 3. Allocation affects effective attention ranges across t, x, y.

Notice **VideoRoPE** focuses on **temporally aligned** frames, whereas M-RoPE attends **broadly** via spatial cues.

Qual **Focused temporal attention**

Qualitative: Temporal & Hallucination

Temporal Allocation

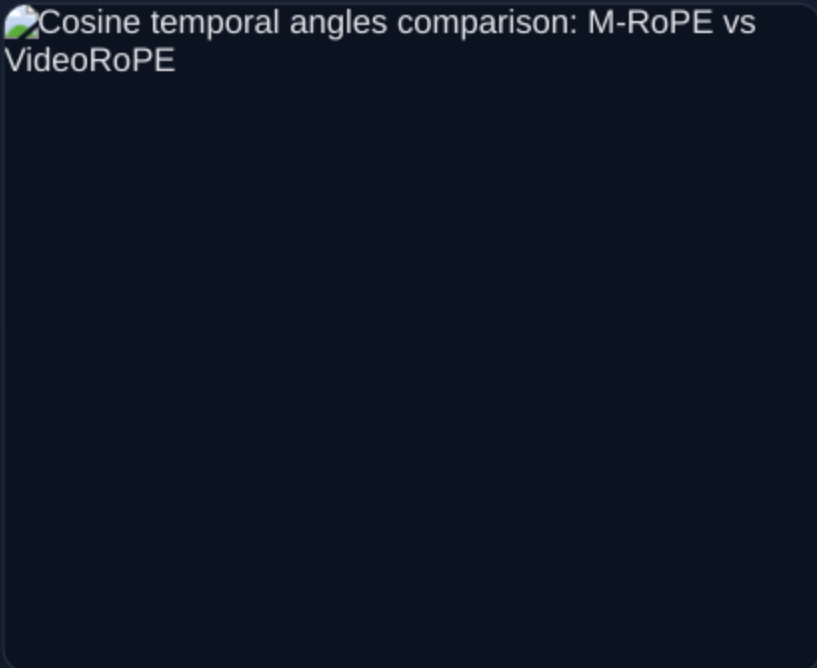
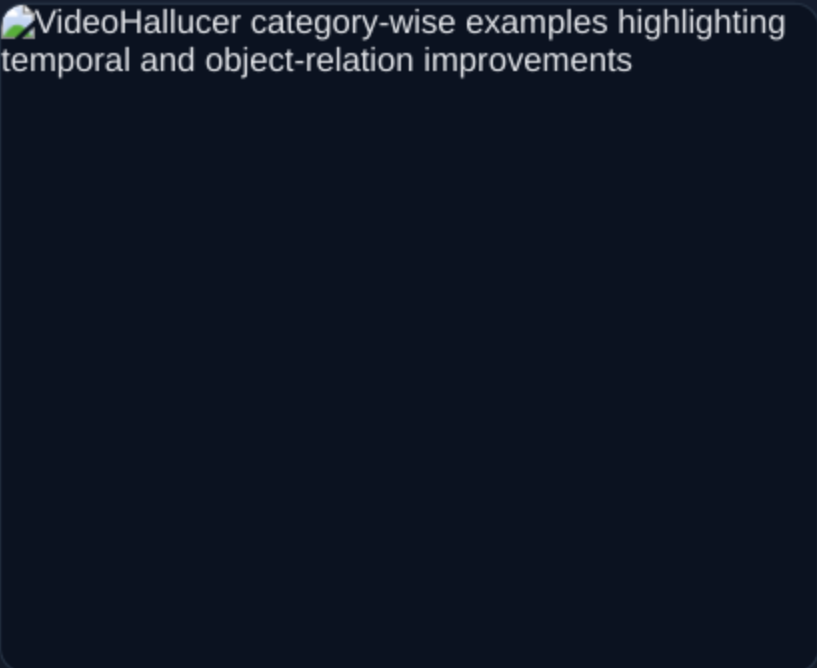


Figure 4. Wider monotonic intervals for t in VideoRoPE mitigate Notice **low-frequency** oscillations reduces **hash collisions**, yielding fewer **hallucinations** in temporal and object-relation queries.

Qual

Video Hallucination



VideoHalluciner. Higher scores in Temporal and Object-Relation yielding fewer **hallucinations** in temporal and object-relation queries.

Why it works